Ej 2

$$\forall t, u :: AB\ a \quad \text{altura } t \geq \text{altura } (\text{zipAB } t\ u)$$

$\forall t :: AB\ a \cdot P(t)$ con

$$P(t) \equiv \forall u :: AB\ a \quad \text{altura } t \geq \text{altura } (\text{zipAB } t\ u)$$

Por inducción estr. en $AB\ a$, si

- $P(\text{Nil})$
- $\forall i, d :: AB\ a.\ r{:}a \cdot \underbrace{P(i) \wedge P(d)}_{HI} \Rightarrow P(\text{Bin } i\ r\ d)$

entonces sabemos $\forall t :: AB\ a \cdot P(t)$

Plan

▷ Caso $P(Nil)$

· ecuaciones directas para un u cualquiera

▷ Caso $P(Bin\ i\ r\ d)$

Sean $i\ r\ d$ cualesquiera

asumimos $P(i) \wedge P(d)$

Qvq $\underline{P(Bin\ i\ r\ d)}$

$\hookrightarrow \forall u : AB\ a \ldots$

Lema $\mathbb{Z}$

1. $\max\ a\ b = \max\ b\ a$
2. $b \geq c \Rightarrow$ $\max\ a\ b \geq \max\ a\ c$
3. $b \geq c \Rightarrow$ $\max\ b\ a \geq \max\ c\ a$

× lema de gen en AB

$u = Nil$ $\qquad$ $u = Bin\ u_i\ u_r\ u_d$

LEMA { $\qquad$ ∅ $\qquad$ {

✓ $\qquad$ ✓

▷ Caso $P(Nil)$

Qvq $\forall u :: ABa \cdot altura\ (Nil) \geq altura\ (zip\ AB\ Nil\ u)$

Sea $u :: ABa$ cualquiera

Qvq

$$altura\ (Nil) \geq altura\ (zip\ AB\ Nil\ u)$$

$$\underset{\{Z_0\}}{\equiv} \quad altura\ (Nil) \geq altura\ (const\ Nil\ u)$$

$$\underset{\{C\}}{\equiv} \quad altura\ Nil \geq altura\ ((\backslash x \to \backslash y \to x)\ Nil\ u)$$

$$\underset{\{\beta\}}{\equiv} \quad altura\ Nil \geq altura\ ((\backslash y \to Nil)\ u)$$

$$\underset{\{\beta\}}{\equiv} \quad altura\ Nil \geq altura\ Nil$$

$\boxed{\checkmark}$ (por $x = x \Rightarrow x \geq x$ $\forall x :: Int$)

(opcional)

$$\underset{\{A\emptyset\}}{\equiv} \quad 0 \geq altura\ Nil$$

$$\underset{\{A\emptyset\}}{\equiv} \quad 0 \geq 0$$

$\boxed{\checkmark}$ por $\mathbb{Z}$

D Caso $P(Bin\ i\ r\ d)$

Sabemos $\underbrace{P(i) \wedge P(d)}_{HI}$ qvq $P(Bin\ i\ r\ d)$

$P(i) \equiv \forall u :: AB\ a \cdot$ altura $i \geq$ altura $(zipAB\ i\ u)$

$P(d) \equiv \forall u :: AB\ a \cdot$ altura $d \geq$ altura $(zipAB\ d\ u)$

pueden ser distintos u

Qvq $P(Bin\ i\ r\ d)$

Qvq $\forall u :: AB\ a$ altura $(Bin\ i\ r\ d) \geq$ altura $((zipAB\ (Bin\ i\ r\ d)\ u)$

Sea $u :: AB\ a$ cualquiera

Qvq altura $(Bin\ i\ r\ d) \geq$ altura $((zipAB\ (Bin\ i\ r\ d)\ u)$

$\underset{\{AL\}}{=} 1 + \max$ (altura i) (altura d) $\geq$ altura $(zipAB\ (Bin\ i\ r\ d)\ u)$

por lema de generación en $AB\ a$. $u = Nil$ o

$u = Bin\ u_i\ u_r\ u_d$

para algún $u_i, u_d :: AB\ a$ $u_r :: a$

▷ Caso $u = \text{Nil}$

... pendiente ...

▷ Caso $u = \text{Bin}\ u_i\ u_r\ u_d$

$$\underbrace{1 + \max\ (\text{altura}\ i)\ (\text{altura}\ d)}_{E1} \geq \text{altura}\ (\text{zipAB}\ (\text{Bin}\ i\ r\ d)\ u)$$

Qvq

$$E1 \geq \text{altura}\ (\text{zipAB}\ (\text{Bin}\ i\ r\ d)\ (\text{Bin}\ u_i\ u_r\ u_d))$$

$$\overset{\equiv}{\{Z_1\}} \quad E1 \geq \text{altura}\ ((\backslash t \to \text{case}\ t\ d \ldots)\ (\text{Bin}\ u_i\ u_r\ u_d))$$

$$\overset{\equiv}{\{B\}\ \text{case}} \quad E1 \geq \text{altura}\ (\text{Bin}\ (\text{zipAB}\ i\ u_i)\ (r, u_r)\ (\text{zipAB}\ d\ u_d))$$

$$\overset{\equiv}{\{A1\}} \quad E1 \geq 1 + \max\ (\text{altura}\ (\text{zipAB}\ i\ u_i))\ (\text{altura}\ (\text{zipAB}\ d\ u_d))$$

$$\overset{\equiv}{\{Z\}} \quad \max\ (\text{altura}\ i)\ (\text{altura}\ d) \geq \max\ (\text{altura}\ (\text{zipAB}\ i\ u_i))\ (\text{altura}\ (\text{zipAB}\ d\ u_d))$$

qvq

$$\max\ (\text{altura } i)\ (\text{altura } d) \geq \max\ (\text{altura } (\text{zipAB } i\ u_i))\ (\text{altura } (\text{zipAB } d\ u_d))$$

Sabemos

$$P(i) \equiv \forall u :: AB\ a \cdot \text{altura } i \geq \text{altura } (\text{zipAB } i\ u)$$

$$P(d) \equiv (\forall u :: AB\ a \cdot \text{altura } d \geq \text{altura } (\text{zipAB } d\ u)$$

pueden ser distintos u

① $\max\ a\ b = \max\ b\ a$

② $b \geq c \Rightarrow \max\ a\ b \geq \max\ a\ c$

③ $b \geq c \Rightarrow \max\ b\ a \geq \max\ c\ a$

Entonces

Por $P(i)$ y ③ en $u = u_i$

$$\max\ (\text{altura } i)\ (\text{altura } d) \geq \max\ (\text{altura } (\text{zipAB } i\ u_i))\ (\text{altura } d) \quad \text{E2}$$

Por $P(d)$ y ② en $u = u_d$

$$\text{E2} \geq \max\ (\text{altura } (\text{zipAB } i\ u_i))\ (\text{altura } (\text{zipAB } d\ u))$$

$\boxed{\checkmark}$

Ej 3

$$
\dfrac{
\dfrac{}{\Gamma \vdash \sigma \vee (\rho \Rightarrow \tau)}\,\mathrm{Ax}
\qquad
\dfrac{\dfrac{}{\Gamma, \sigma \vdash \sigma}\,\mathrm{Ax}}{\Gamma, \sigma \vdash \sigma \vee \tau}\,\vee i_1
\qquad
\dfrac{\dfrac{\dfrac{}{\Gamma, \rho \Rightarrow \tau \vdash \rho \Rightarrow \tau}\,\mathrm{Ax} \qquad \dfrac{}{\Gamma, \rho \Rightarrow \tau \vdash \rho}\,\mathrm{Ax}}{\Gamma, \rho \Rightarrow \tau \vdash \tau}\,\Rightarrow e}{\Gamma, \rho \Rightarrow \tau \vdash \sigma \vee \tau}\,\vee i_2
}{
\dfrac{\dfrac{\Gamma, \rho, (\sigma \vee (\rho \Rightarrow \tau)) \vdash \sigma \vee \tau}{\rho \vdash (\sigma \vee (\rho \Rightarrow \tau)) \Rightarrow (\sigma \vee \tau)}\,\Rightarrow i}{\vdash \rho \Rightarrow ((\sigma \vee (\rho \Rightarrow \tau)) \Rightarrow (\sigma \vee \tau))}\,\Rightarrow i
}\,\vee e
$$

Ej 4

$\tau ::= \cdots \mid \mathrm{Cola}\ \tau$

$M ::= \cdots \mid \langle\rangle_\tau \mid M \bullet M \mid \text{próx}(M) \mid \text{des}(M)$
$\mid \text{case } M \text{ of } \langle\rangle \rightsquigarrow M;\ c \bullet x \rightsquigarrow M$

$$\frac{}{\Gamma \vdash \langle\rangle_\tau : \mathrm{Cola}\ \tau}\ \text{T-VACÍA} \qquad \frac{\Gamma \vdash M : \mathrm{Cola}\ \tau \quad \Gamma \vdash N : \tau}{\Gamma \vdash M \bullet N : \mathrm{Cola}\ \tau}\ \text{T-AGR.}$$

$$\frac{\Gamma \vdash M : \mathrm{Cola}\ \tau}{\Gamma \vdash \text{próx}(M) : \tau}\ \text{T-PRÓX} \qquad \frac{\Gamma \vdash M : \mathrm{Cola}\ \tau}{\Gamma \vdash \text{des}(M) : \mathrm{Cola}\ \tau}\ \text{T-DES}$$

$$\frac{\Gamma \vdash M : \mathrm{Cola}\ \tau \quad \Gamma \vdash N : \gamma \quad \Gamma, c : \mathrm{Cola}\ \tau, x : \tau \vdash O : \gamma}{\Gamma \vdash \text{case } M \text{ of } \langle\rangle \rightsquigarrow N;\ c \bullet x \rightsquigarrow O : \gamma}\ \text{T-CASE}$$

$V ::= \cdots \mid \langle\rangle_\tau \mid V \bullet V$

$$V ::= \cdots \mid \langle\rangle_\tau \mid V \bullet V$$

Reglas de congruencia

$$\frac{M \to M'}{M \bullet N \to M' \bullet N}\ \text{E-IZQ}\bullet \qquad \frac{N \to N'}{V \bullet N \to V \bullet N'}\ \text{E-DER}\bullet$$

$$\frac{M \to M'}{\text{próx}(M) \to \text{próx}(M')}\ \text{E-PROX} \qquad \frac{M \to M'}{\text{des}(M) \to \text{des}(M')}\ \text{E-DES}$$

$$\frac{M \to M'}{\text{case } M \text{ of } \langle\rangle \rightsquigarrow N;\ c \bullet x \rightsquigarrow O \to \text{case } M' \text{ of } \langle\rangle \rightsquigarrow N;\ c \bullet x \rightsquigarrow O}\ \text{E-CASE}$$

Reglas de cómputo

$$\frac{}{\text{próx}(\langle\rangle_\tau \bullet V_1) \to V_1}\ \text{E-PROXV} \qquad \frac{}{\text{próx}((V_1 \bullet V_2) \bullet V_3) \to \text{próx}(V_1 \bullet V_2)}\ \text{E-PROX2}$$

$$\frac{}{\text{des}(\langle\rangle_\tau \bullet V_1) \to \langle\rangle_\tau}\ \text{E-DESV} \qquad \frac{}{\text{des}((V_1 \bullet V_2) \bullet V_3) \to \text{des}(V_1 \bullet V_2) \bullet V_3}\ \text{E-DES2}$$

$$\frac{}{\text{case } \langle\rangle_\tau \text{ of } \langle\rangle \rightsquigarrow M;\ c \cdot x \rightsquigarrow N \longrightarrow M}\ \text{E-CASEV}$$

$$\frac{}{\text{case } V_1 \cdot V_2 \text{ of } \langle\rangle \rightsquigarrow M;\ c \cdot x \rightsquigarrow N \rightarrow N\{c := V_1, x := V_2\}}\ \text{E-CASE2}$$

$$\text{case } \underbrace{(\langle\rangle_{Nat} \cdot \underline{1}) \cdot \underline{0}} \text{ of } \langle\rangle \rightsquigarrow \underbrace{\text{próximo}(\langle\rangle_{Bool})};\ c \cdot x \rightsquigarrow \underbrace{\text{iszero}(x)}$$

$$\xrightarrow[\text{E-CASE2}]{} \text{iszero}(x)\{x := \underline{0}\} = \text{iszero}(\underline{0}) \xrightarrow[\text{EISZERO}_0]{} \text{True}$$

$$\{c := \langle\rangle_{Nat} \cdot \underline{1}\}$$

$$\text{último}_\tau := \lambda c : \text{Cola}\,\tau.\ \text{case } c \text{ of } \langle\rangle \rightsquigarrow \text{próximo}(\langle\rangle_\tau);\ k \cdot x \rightsquigarrow x$$

$$\vdash \text{último}_\tau : \text{Cola}\,\tau \to \tau$$

$$M ::= \ldots \mid \bot_\sigma$$

$$\frac{}{\Gamma \vdash \bot_\sigma : \sigma}\ \text{T-BOTTOM}$$